

The ACM Multimedia 2023 Deep Video Understanding Grand Challenge

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ABSTRACT

This is the overview paper for the Deep Video Understanding (DVU) Grand Challenge. In recent years, a growing trend towards working on understanding videos (in particular movies) to a deeper level started to motivate researchers working in multimedia and computer vision to present new approaches and datasets to tackle this problem. This is a challenging research area which aims to develop a deep understanding of the relations which exist between different individuals and entities in movies using all available modalities such as video, audio, text and metadata. The aim of this grand challenge is to foster innovative research in this new direction and to provide benchmarking evaluations to advance technologies in the deep video understanding community.

CCS CONCEPTS

• **Information systems** → *Multimedia content creation*; **Multimedia and multimodal retrieval**.

KEYWORDS

Video Understanding; Multimedia; Multimodal Interaction; Information Retrieval; Video Ontology

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1 INTRODUCTION

Deep video understanding is a difficult task which requires systems to develop a deep analysis and understanding of the relationships between different entities in video, to use known information to reason about other, more hidden information, and to populate a knowledge graph (KG) representation with all acquired information

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[2]. To work on this task, a system should take into consideration all available modalities (speech, image/video, and in some cases text). The aim of this challenge series is to push the limits of multimodal extraction, fusion, and analysis techniques to address the problem of analyzing long duration videos holistically and extracting useful knowledge to utilize it in solving different types of queries. The target knowledge includes both visual and non-visual elements. As videos and multimedia data are getting more and more popular and usable by users in different domains and contexts, the research, approaches and techniques we aim to be applied in this Grand Challenge will be very relevant in the coming years and near future.

This is the fourth iteration in the series of Deep Video Understanding Grand Challenges at ACM Multimedia. In our first challenge, we focused on video understanding at the whole movie level, such as the relations between different characters and entities in the movie. In the second of these Grand Challenges, we extended video understanding to a deeper level and introduced queries at the scene-level. These scene-level queries focused on finding the unique scene given a set of interactions, finding the next or previous interaction between two characters, matching the scene with a textual description, and scene sentiment classification. In the third year of the DVU challenge [3], we introduced an Explainable AI component, in which participating teams were asked to submit evidence for their answers to scene-level queries. This evidence was in the form of a start and end times of a short video clip from the scene in which evidence of the queried interaction existed, or in the movie-level queries by providing a scene number for evidence of the queried relation. In this fourth year of the Deep Video Understanding Grand Challenge, we are introducing a new robustness sub-task that systems can choose to submit solutions to. To measure the robustness of the multi-modal systems participating in the challenge this year, we will provide a secondary version of the testing dataset after introducing various types of perturbations and corruptions observed in real-world multi-modal data. This will allow teams and organizers to measure how much system performance can get affected by the introduced noise.

2 RELATED WORK

There has been efforts to encourage research in high level video understanding such as the "MovieQA" [8] and "The Large Scale Movie Description Challenge" [7]. However these tasks revolve around

isolated visual concepts retrieval and not about testing systems for their overall understanding of entities, relations and events within the video/movie. Early visions of this kind of work [4] proposed to use visual and audio descriptors, in addition to employing semantic analysis and linking with external knowledge sources in order to populate a knowledge graph. A large-scale dataset of corresponding movie trailers, plots, posters, and metadata was developed by [1] who study the effectiveness of visual, audio, text, and metadata-based features for predicting high-level information about movies such as their genre or estimated budget. One of the most latest research work closely related to the DVU challenge is [5] which focus on complex question answering on long duration movies. However the dataset is not publicly available.

3 DATASET

The Deep Video Understanding Training Set described in Table 1 consists of 19 movies with total duration of about 25 hours¹. This includes 14 CC licensed movies and 5 movies which are licensed from KinoLorberEdu². This training set has been annotated by human assessors and final ground truth, both at the overall movie level (Ontology of relations, entities, actions & events, Knowledge Graph, and names and images of all main characters), and the individual scene level (Ontology of locations, people/entities, interactions and their order between people, sentiments, and text summary) has been provided for the training set to participating researchers for training and development of their systems. In summary, we hired 5 annotators in addition to a summer student and on average each movie took about 20 hours of work to annotate both movie and scenes. A sample from a scene-level knowledge graph annotation can be seen in Figure 2. For more detailed information about the annotation framework please refer to our paper [6].

The DVU Test Set described in Table 2 contains 5 movies licensed from KinoLorberEdu platform with total duration of about 7.5 hours. Participants were required to complete a data access form in order to access these movies. The testing set was fully annotated by human annotators to the same degree as the training set. A set of queries, described in more details in Section 3, were then automatically extracted from human annotations and released to participants, along with the set of movies and annotated images of the movie characters identified during annotation.

More information about movies' genres and duration are provided in Tables 1 and 2.

4 ANNOTATION

Human assessors annotated each movie of the full DVU dataset. Full movies were annotated to a Knowledge Graph (KG) indicating the relationships and connections between every major character, entity, and concept in the movie. Images of each character and entity were also provided. Figure 1 shows an example of a movie-level KG. Following this, every scene within each movie was also annotated to a scene-level KG indicating the locations and characters within each scene, at least one sentiment label for that scene, non-neutral

Movie	Genre	Duration
Honey	Romance	86 min
Let's Bring Back Sophie	Drama	50 min
Nuclear Family	Drama	28 min
Shooters	Drama	41 min
Spiritual Contact		
The Movie	Fantasy	66 min
Super Hero	Fantasy	18 min
The Adventures of Huckleberry Finn	Adventure	106 min
The Big Something	Comedy	101 min
Time Expired	Comedy / Drama	92 min
Valkaama	Adventure	93 min
Bagman	Drama / Thriller	107 min
Manos	Horror	73 min
Road to Bali	Comedy / Musical	90 min
The Illusionist	Adventure / Drama	109 min
Calloused Hands	Drama	92 min
Chained For Life	Comedy / Drama	88 min
Liberty Kid	Drama	88 min
Like Me	Horror / Thriller	79 min
Losing Ground	Comedy / Drama	81 min

Table 1: The Full DVU Training Set

Movie	Genre	Duration
Archipelago	Drama	114 min
Bonneville	Drama	93 min
Heart Machine	Drama	85 min
Littlerock	Drama	82 min
Memphis	Drama	79 min

Table 2: The Full DVU Testing Set

mental states of characters, the interactions between characters, and the ordering of the interactions as they happened in that scene. Figure 2 shows an example of a scene-level KG.

5 QUERIES

5.1 Movie-Level

5.1.1 Fill in the graph space. Fill in spaces in the Knowledge Graph (KG). Given the listed relationships, events or actions for certain nodes, where some nodes are replaced by variables X, Y, etc., solve for X, Y etc. Example of The Simpsons: X Married To Marge. X Friend Of Lenny. Y Volunteers at Church. Y Neighbor Of X. Solution for X and Y in that case would be: X = Homer, Y = Ned Flanders.

5.1.2 Question Answering. This query type represents questions on the resulting KG, including actions and events, of the movies in the described dataset. For example, we may ask "How many children does Person A have?", in which case participating researchers should count the "Parent Of" relationships Person A has in the Knowledge Graph. These are multiple choice questions.

¹<https://www-nlpir.nist.gov/projects/trecvid/dvu/dvu.development.dataset/>

²<https://www.kinolorber.com/>

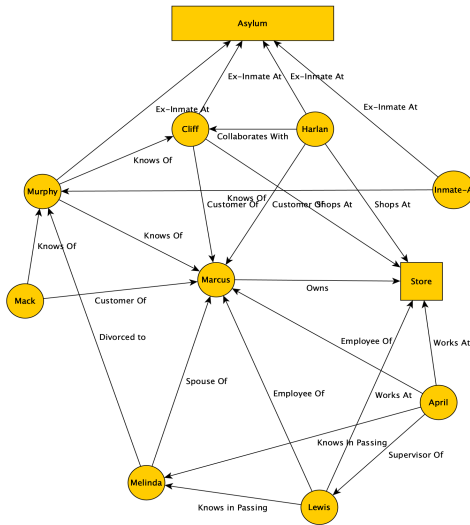


Figure 1: Movie-level KG Sample

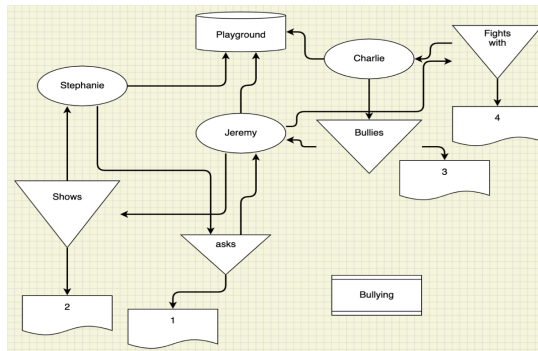


Figure 2: Scene-level KG Sample

5.2 Scene-Level

5.2.1 Find the Unique Scene. Given a full, inclusive list of interactions, unique to a specific scene in the movie, teams should find which scene this is.

5.2.2 Fill in the graph space. Find the person in a specific scene with the following attributes and interactions with others. Participating teams will be given a scene number, a list of person attributes, and a list of interactions to and from other people. Teams should find the only person in that scene with those attributes and interactions.

5.2.3 Find next or previous interaction. Given a specific scene and a specific interaction between person X and person Y, participants will be asked to return either the previous interaction or the next interaction, in either direction, between person X and Person Y. This can be specifically the next or previous interaction within the same scene, or over the entire movie. These will be multiple choice questions selected from a list of possible interactions, only one of which will be correct.

5.2.4 Find the 1-to-1 relationship between scenes and natural language descriptions. Given a set of scenes, and a set of natural language descriptions of movie scenes, match the correct natural language description for each scene.

5.2.5 Classify scene sentiment from a given scene. Given a specific movie scene and a set of possible sentiments, classify the correct sentiment label for each given scene.

6 METRICS

6.1 Movie-Level

6.1.1 Fill in the graph space. Results will be treated as ranked list of result items per each unknown variable and the Reciprocal Rank score will be calculated per unknown variable and Mean Reciprocal Rank (MRR) per query.

6.1.2 Question Answering. Scores for this query will be calculated by the number of Correct Answers / number of Total Questions.

6.2 Scene-Level

6.2.1 Find the Unique Scene. Results will be treated as ranked list of result items per each unknown variable and the Reciprocal Rank score will be calculated per unknown variable and Mean Reciprocal Rank (MRR) per query.

6.2.2 Fill in the graph space. Results will be treated as ranked list of result items per each unknown variable and the Reciprocal Rank score will be calculated per unknown variable and Mean Reciprocal Rank (MRR) per query.

6.2.3 Find next or previous interaction. Scores for this query will be calculated by the number of Correct Answers / number of Total Questions.

6.2.4 Find the 1-to-1 relationship between scenes and natural language descriptions. Scores for this query will be calculated by the number of Correct Answers / number of Total Questions.

6.2.5 Classify scene sentiment from a given scene. Scores for this query will be calculated by the number of Correct Answers / number of Total Questions.

7 RUN SUBMISSION TYPES

Teams are allowed to submit to either the Movie-level queries only, the scene-level queries only, or both the movie and scene queries. For movie-level queries, teams can choose to submit results to either or both groups of:

- Group 1 - Question 1 (fill in the graph)
- Group 2 - Question 2 (question answering)

For scene-level queries, teams can choose to submit results to either or both groups of:

- Group 1 - Questions 1, 2, & 3 (all related to interactions between characters)
- Group 2 - Questions 4 & 5 (matching of scenes and their descriptions in text, and sentiment classification)

Submissions for both movie-level and scene-level queries were in XML format. This allowed for the automatic scoring of solutions to the challenge. Below is an example XML solution to a scene-level

Group	Rank 1	Rank 2	Rank 3	Rank 4
Group 1	STARS	WHU_NERCMS	MINE-MM	MAGUS.LFYT
Group 2	WHU_NERCMS	MAGUS.LFYT	MINE-MM	STARS

Table 3: Movie-Level Rank Table

Group	Rank 1	Rank 2	Rank 3	Rank 4
Group 1	WHU_NERCMS	MAGUS.LFYT	MINE-MM	DSSC
Group 2	MAGUS.LFYT	WHU_NERCMS	DSSC	MINE-MM

Table 4: Scene-Level Rank Table

previous interaction question from a previous challenge. This indicates the submitted answer of “watches” for the type “Interaction”.

```
<DeepVideoUnderstandingTopicResult id="2" question="4">
  <item answer="watches" type="Interaction"/>
</DeepVideoUnderstandingTopicResult>
```

8 PARTICIPATING TEAMS AND RANKINGS

A total of 5 unique teams participated in this year challenge. In scene-level queries, 3 teams participated in Group 1 and 4 teams in Group 2. For the movie-level queries, 4 teams participated in Group 1 and 4 teams in Group 2. Only 1 team (MAGUS.LFYT) submitted results for the optional robustness in multimodal task. Tables 3 and 4 show final ranks of teams for movie and scene level queries. Average scores in movie-level queries group 1 from all submissions ranged between 0.36 to 0.74, while for group 2 ranged between 0.38 to 0.44. For scene-level queries group 1, scores ranged between 0 to 0.89, while for group 2, ranged between 0.13 to 0.86. As general observations, we noticed that for movie-level questions, group 2 scores are less than group 1 indicating that open domain question answering has been harder. On the other hand for scene-level queries, group 2 (sentiment classification and scene to text summary) scores are higher than group 1 indicating graph analysis and interaction identification across scenes are more challenging. For more detailed analysis for system approaches, we encourage readers to refer back to participants’ papers.

9 CONCLUSIONS

Running the DVU grand challenge during the last 4 years has been very exciting and challenging at the same time. It is exciting to see new research and techniques trying to address the various research questions posed by the organizers, and challenging due to the labor and cost involved in identifying reliable data collection sources, human annotation, and reviewing generated queries. We

hope that attendees and readers of systems’ papers and sessions find a good diversity of research presented at this challenge, and we hope that we have encouraged and inspired further work in this new direction.

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